

**Is there an association between age at first live birth and current
socioeconomic status among U.S. women?**

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Abstract

Background:

Age at first live birth is an important determinant of women's long-term socioeconomic outcomes. Early childbearing has been associated with lower educational attainment and reduced economic stability, yet fewer studies have examined how reproductive timing relates to current socioeconomic status using nationally representative data.

Methods:

This cross-sectional study used data from the National Health and Nutrition Examination Survey (NHANES) 2009-2012. The analytic sample included female participants with complete data on age at first live birth, poverty income ratio, median household income, and selected covariates. Descriptive statistics and bivariate analysis were conducted across categorized age at first birth. Associations were evaluated using simple and multivariable linear regression models, as well as logistic regression models. Statistical significance was assessed at 0.05.

Results:

Later age at first live birth was consistently associated with more favorable socioeconomic outcomes. In unadjusted models, each additional year in Age1stBaby was associated with higher poverty income ratio and higher household income. Logistic regression showed decreased odds of living below the poverty threshold with increasing Age1stBaby. Associations remained significant but attenuated after adjustment for covariates.

Discussion: Later age at first birth is associated with improved socioeconomic status among U.S. women. These findings support the life course perspective and highlight the importance of reproductive timing in shaping long-term economic outcomes. Public health

interventions addressing reproductive health and social determinants may help reduce socioeconomic disparities as well as inform maternal and child health (MCH) policies.

Introduction

Background

In the United States, roughly 20 percent of women give birth before age 20, and early childbearing can have lasting effects on economic stability and health (**Levine2009?**). The timing of first childbirth is a critical public health concern, as socioeconomic disadvantage strongly shapes access to education, resources, and overall well-being (**Moore1993?**; **Khan2023?**).

The National Health and Nutrition Examination Survey (NHANES), conducted by the U.S. National Center for Health Statistics, collects nationally representative data on health, nutrition, and demographics. This data provides a unique opportunity to examine how reproductive timing relates to current socioeconomic status (**Council1987?**). Data from 2009 – 2012 cycles, including approximately 5,000 participants, allow for robust population-level analyses to inform public health interventions and policy.

Socioeconomic status (SES), measured by poverty income ratio (PIR), is the primary outcome of interest. Early childbearing is consistently associated with lower education, reduced income, and larger family size, whereas delayed childbearing is generally linked to higher educational attainment and economic stability (**Câmara2015?**; **Gigante2019?**; **MonroyDeVelasco1982?**). Social determinants of health including race/ethnicity, education, and access to services interact with reproductive timing to shape these outcomes (**Jakubowski2023?**; **Khan2023?**). This correlation highlights a significant public health perspective called the life course perspective often addressed in maternal and child health interventions. The life course perspective uses the ideology that an individual's overall health and well-being is a lifelong process shaped by all the lived experiences encountered throughout one's lifespan (**Kirby2022?**).

Although many studies explore socioeconomic predictors of adolescent pregnancy, fewer examine how the timing of first childbirth predicts current economic outcomes (**Levine2009?; Moore1993?**). This study addresses that gap by asking: “Is there an association between age at first live birth and current socioeconomic status among U.S. women?” It is hypothesized that women who had their first child at an older age are less likely to live below the poverty threshold, highlighting the long-term socioeconomic consequences of reproductive timing.

Purpose and Objectives

The purpose of this study was to examine the association between age at first live birth and current socioeconomic status among U.S. women using the NHANES 2009-2012 data, in hope to help inform maternal and child health (MCH) policies.

- To assess the relationship between Age1stBaby and poverty income ratio (Poverty) as a continuous measure of socioeconomic status.
- To assess the relationship between Age1stBaby and median household income (HHIncomeMid) as a continuous measure of socioeconomic status.

Hypotheses

Null Hypothesis: There is no association between age at first live birth and socioeconomic status among U.S. women.

Alternative Hypothesis: There is an association between age at first live birth and socioeconomic status among U.S. women.

Methods

Inclusion and Exclusion Criteria

This study included all female participants of the National Health and Nutrition Survey (NHANES), years 2009 to 2012, with completed data for age at first live birth, poverty income ratio, household income, race/ethnicity, education, marital status, and current age. Male participants and all female participants with missing data for any of the variable required in this analysis were excluded.

Data Preparation and Management

Data preparation and management were conducted in R using tidyverse. The NHANES dataset was filtered to include female participants only. All participants with incomplete or missing values for variables used within this statistical analysis were removed using the `drop_na()` function. A derived categorical variable, `Age1stBabyDecade` was created from age at first birth using predefined decades intervals (<20, 20-29, 30-39, 40-49, 50-59, 60-69, 70+). For the purpose of this statistical analysis, all unused factor levels were removed so that tables and figures would appear cleaner. Additionally, a binary variable, `poverty_status` was created based on the federal poverty threshold, where poverty income ratio (PIR) < 1 indicated below poverty status. All categorical variables were formatted as factors.

Study Variables

Table 1. Data Dictionary of Variables Used in NHANES Analysis

Variable	Description	Type	Coding
Gender	Participant gender	Categorical	Male, Female
Age1stBaby	Age at first live birth	Continuous	Years

Variable	Description	Type	Coding
Age1stBabyDecade	Age at first birth (grouped)	Categorical	<20, 20-29, 30-39, 40-49, 50-59, 60-69, 70+
Race1	Race/ethnicity	Categorical	NHANES categories
Poverty	Poverty Income Ratio (PIR)	Continuous	Ratio (0.0 - 5.0+)
poverty_status	Poverty status indicator	Binary	1 = Below poverty, 0 = Above poverty
Age	Current age of participant	Continuous	Years
AgeDecade	Current age of participant (grouped)	Categorical	<20, 20-29, 30-39, 40-49, 50-59, 60-69, 70+
HHIncomeMid	Household Income (midpoint estimate)	Continuous	United States Dollar (USD)
MaritalStatus	Current marital status	Categorical	NHANES categories
Education	Educational attainment	Categorical	NHANES categories

Note. Data from National Health and Nutrition Examination Survey (NHANES). PIR = Poverty Income Ratio. Age at first birth categorized into decade-based groups for analysis.

The independent variable for this study was age at first live birth (Age1stBaby), measured in years as a continuous variable. For descriptive and stratified analyses, age at first birth was converted into categorical decade-based groups (Age1stBabyDecade). The dependent variable for this study was socioeconomic status. Poverty income ratio (Poverty), a continuous variable, was used at the primary indicator for socioeconomic status. Median household income (HHIncomeMid), a continuous variable, was utilized in the statistical analysis as a secondary socioeconomic status indicator. Poverty income ratio was converted into a binary variable for stratified analyses and logistic models, coded as poverty_status indicating whether the participants is above or below federal poverty threshold. The covariates included current age (continuous), race/ethnicity (categorical), marital status (categorical), and educational attainment (categorical).

Statistical Analysis

This study included a cross-sectional secondary data analysis using publicly available nationally representative data from the National Health and Nutrition Examination Survey

(NHANES) from 2009 to 2012. Data was obtained from the NHANES dataset available through RStudio. The analytic sample included female participants with complete data on reproductive history, socioeconomic indicators, and selected covariates. All statistical tests were assessed with a significance level of 0.05. Analyses were conducted using the latest 2026 version of RStudio.

Descriptive statistics were generated using gtsummary package. Continuous variables were summarized using means and standard deviations, while categorical variables were summarized using frequencies and percentages.

Bivariate analyses compared socioeconomic outcomes across age at first birth categories using Kruskal-Wallis tests for continuous variables and Pearson's chi-square tests for categorical variables. Two simple linear regression models were conducted, assessing the association between age at first birth and poverty income ratio (PIR) as well as median household income.

Multivariable linear regression models were constructed to adjust for age, race/ethnicity, marital status, and education. An interaction model was fitted to evaluate whether the association between age at first birth and poverty income ratio differed by race/ethnicity.

Logistic regression models were used to estimate the odds of living below the poverty threshold (poverty_status) as a function of age at first birth. An adjusted logistic regression model was used to adjust for covariates.

Results

Descriptive Statistics

Characteristic	Overall (N = 1,737)¹
Current Age	54 (16)
Race/Ethnicity	
<i>Black</i>	178 (10%)
<i>Hispanic</i>	96 (5.5%)
<i>Mexican</i>	120 (6.9%)
<i>White</i>	1,252 (72%)
<i>Other</i>	91 (5.2%)
Education	
<i>8th Grade</i>	94 (5.4%)
<i>9 - 11th Grade</i>	239 (14%)
<i>High School</i>	409 (24%)
<i>Some College</i>	549 (32%)
<i>College Grad</i>	446 (26%)
Marital Status	
<i>Divorced</i>	226 (13%)
<i>LivePartner</i>	99 (5.7%)
<i>Married</i>	1,045 (60%)
<i>NeverMarried</i>	72 (4.1%)
<i>Separated</i>	58 (3.3%)
<i>Widowed</i>	237 (14%)
Household Income	54,780 (32,623)
Poverty Income Ratio	2.84 (1.64)
Age at First Birth	23 (5)

Characteristic	Overall (N = 1,737) ¹
Poverty Status	
<i>Above Poverty</i>	1,448 (83%)
<i>Below Poverty</i>	289 (17%)

¹Note. Values are presented as mean (SD) for continuous variables and n (%) for categorical variables.

A total of 1737 female participants were included in the final analytic sample after exclusion of males and observations missing data. The mean age of participants was 53.5 years (SD = 15.7). Overall, descriptive characteristics are presented in **Table 2**. The mean age at first live birth was 22.7 years. The mean poverty income ratio (PIR) was 2.84.

Figure 1. Distribution of Poverty Income Ratio by Age at First Birth Decade (NHANES)

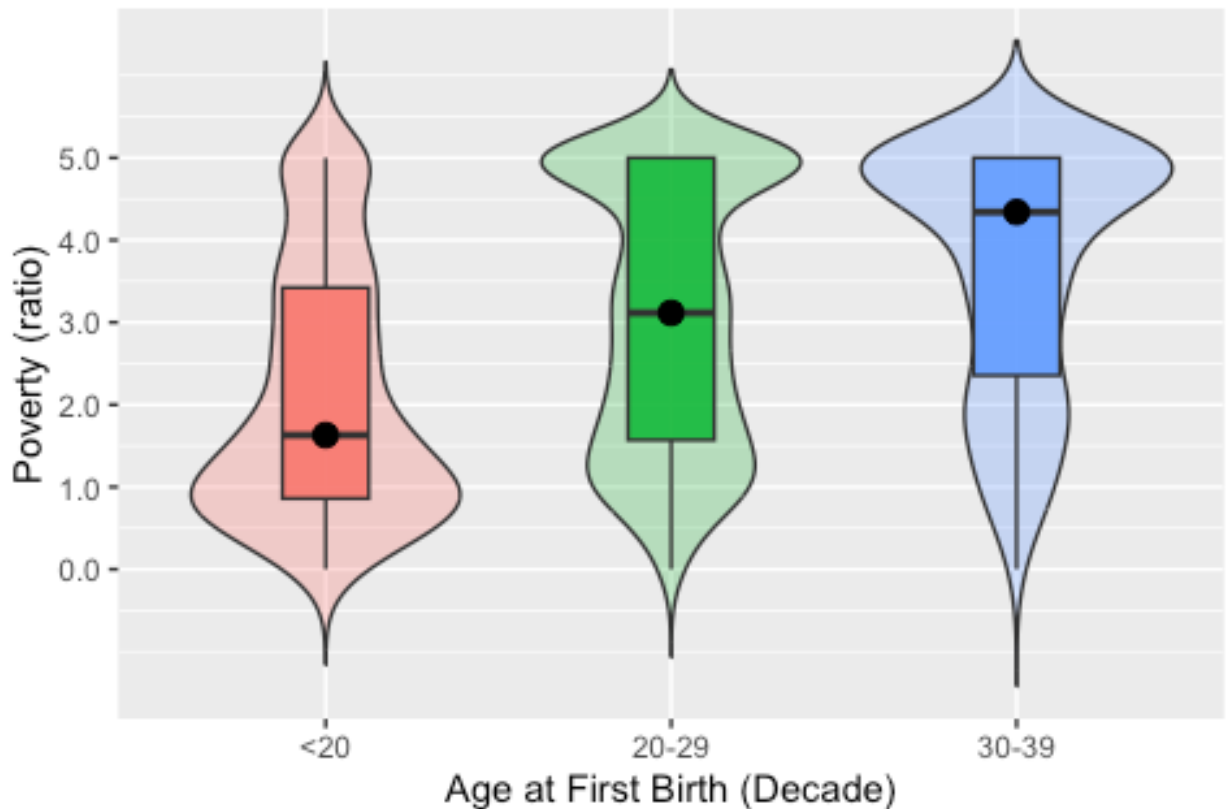


Figure 1 depicts the distribution of poverty income ratio (PIR) across age at first birth categories. The median PIR increases across age categories with the older ages (20-29, 30-39) having showing a higher median PIR. The figure also shows the distribution of PIR values across all age groups varying in ranges and interquartile spreads.

Bivariate Analysis

Characteristic	<20 N = 500 ¹	20-29 N = 1,062 ¹	30-39 N = 175 ¹	p-value ²
Current Age	52 (17)	55 (15)	51 (12)	<0.001
Race/Ethnicity				<0.001
<i>Black</i>	87 (17%)	86 (8.1%)	5 (2.9%)	
<i>Hispanic</i>	27 (5.4%)	58 (5.5%)	11 (6.3%)	
<i>Mexican</i>	62 (12%)	55 (5.2%)	3 (1.7%)	
<i>White</i>	312 (62%)	793 (75%)	147 (84%)	
<i>Other</i>	12 (2.4%)	70 (6.6%)	9 (5.1%)	
Education				<0.001
<i>8th Grade</i>	52 (10%)	41 (3.9%)	1 (0.6%)	
<i>9 - 11th Grade</i>	124 (25%)	109 (10%)	6 (3.4%)	
<i>High School</i>	153 (31%)	244 (23%)	12 (6.9%)	
<i>Some College</i>	145 (29%)	371 (35%)	33 (19%)	
<i>College Grad</i>	26 (5.2%)	297 (28%)	123 (70%)	
Marital Status				<0.001
<i>Divorced</i>	76 (15%)	132 (12%)	18 (10%)	
<i>LivePartner</i>	58 (12%)	36 (3.4%)	5 (2.9%)	
<i>Married</i>	224 (45%)	686 (65%)	135 (77%)	
<i>NeverMarried</i>	46 (9.2%)	25 (2.4%)	1 (0.6%)	
<i>Separated</i>	18 (3.6%)	32 (3.0%)	8 (4.6%)	
<i>Widowed</i>	78 (16%)	151 (14%)	8 (4.6%)	

Characteristic	<20 N = 500 ¹	20-29 N = 1,062 ¹	30-39 N = 175 ¹	p-value ²
Household Income	40,165 (28,663)	58,409 (32,145)	74,514 (29,590)	<0.001
Poverty Income Ratio (PIR)	2.10 (1.49)	3.05 (1.60)	3.67 (1.48)	<0.001
Age at First Birth	17 (1)	24 (3)	32 (2)	<0.001
Poverty Status				<0.001
<i>Above Poverty</i>	341 (68%)	942 (89%)	165 (94%)	
<i>Below Poverty</i>	159 (32%)	120 (11%)	10 (5.7%)	

¹Mean (SD); n (%)

²Kruskal-Wallis rank sum test; Pearson's Chi-squared test

Table 3 presents participant characteristics stratified by age at first live birth. Significant differences were observed across age at first birth categories for socioeconomic indicators.

Poverty income ratio differed significantly across groups ($p < 0.05$), with later age at first birth associated with higher mean PIR values. Similarly, household income (HHIncomeMid) increased across age at first birth categories.

Figure 1 illustrated the distribution of poverty income ratio across age at first birth decades. Participants who had their first child at younger ages showed lower median PIR values compared to those with later births.

Regression Analysis

Figure 2. Association Between Age at First Birth and Poverty Income Ratio (NHANES)

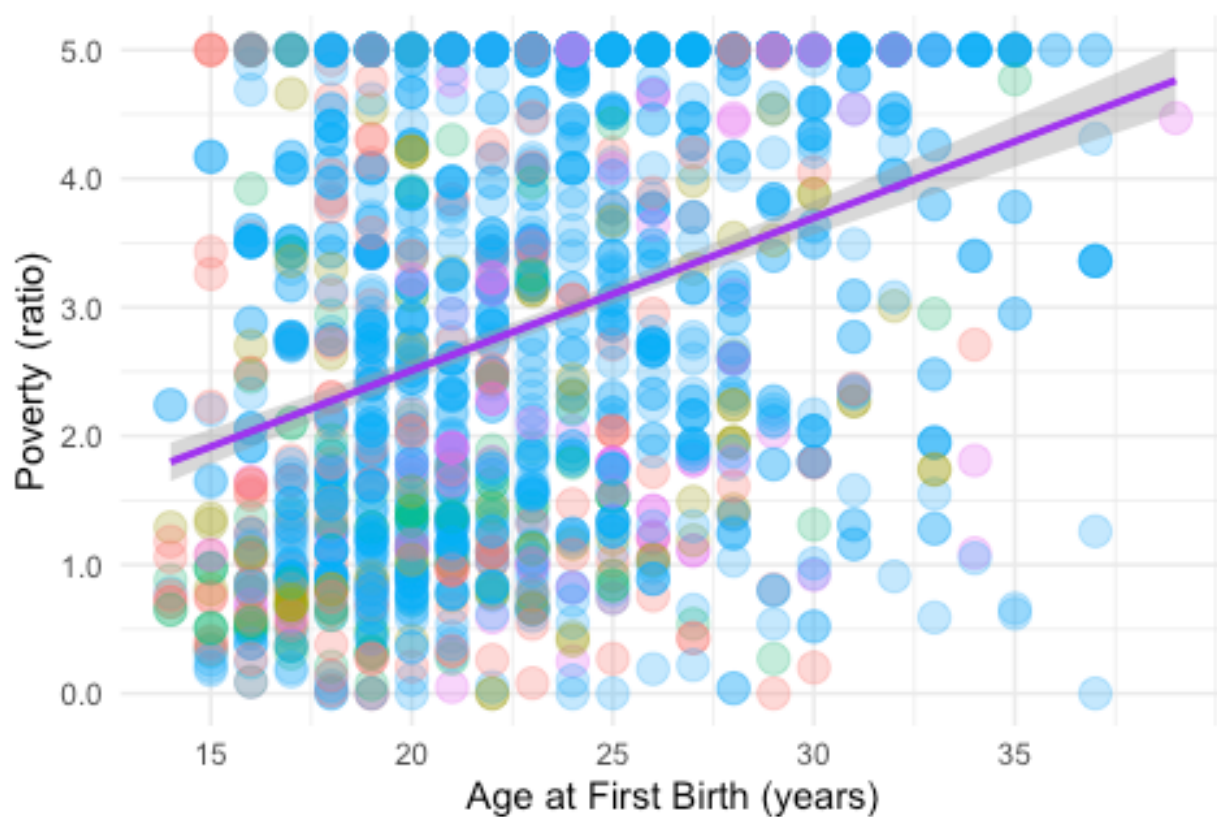
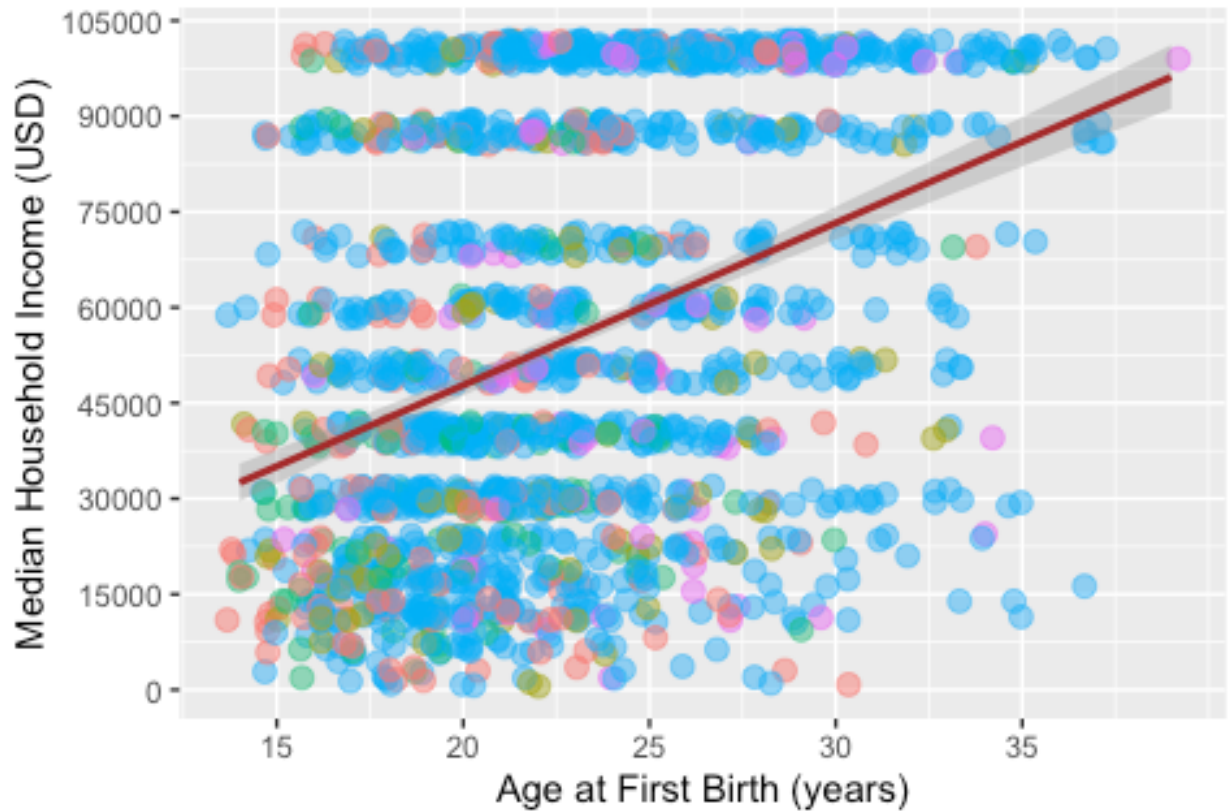


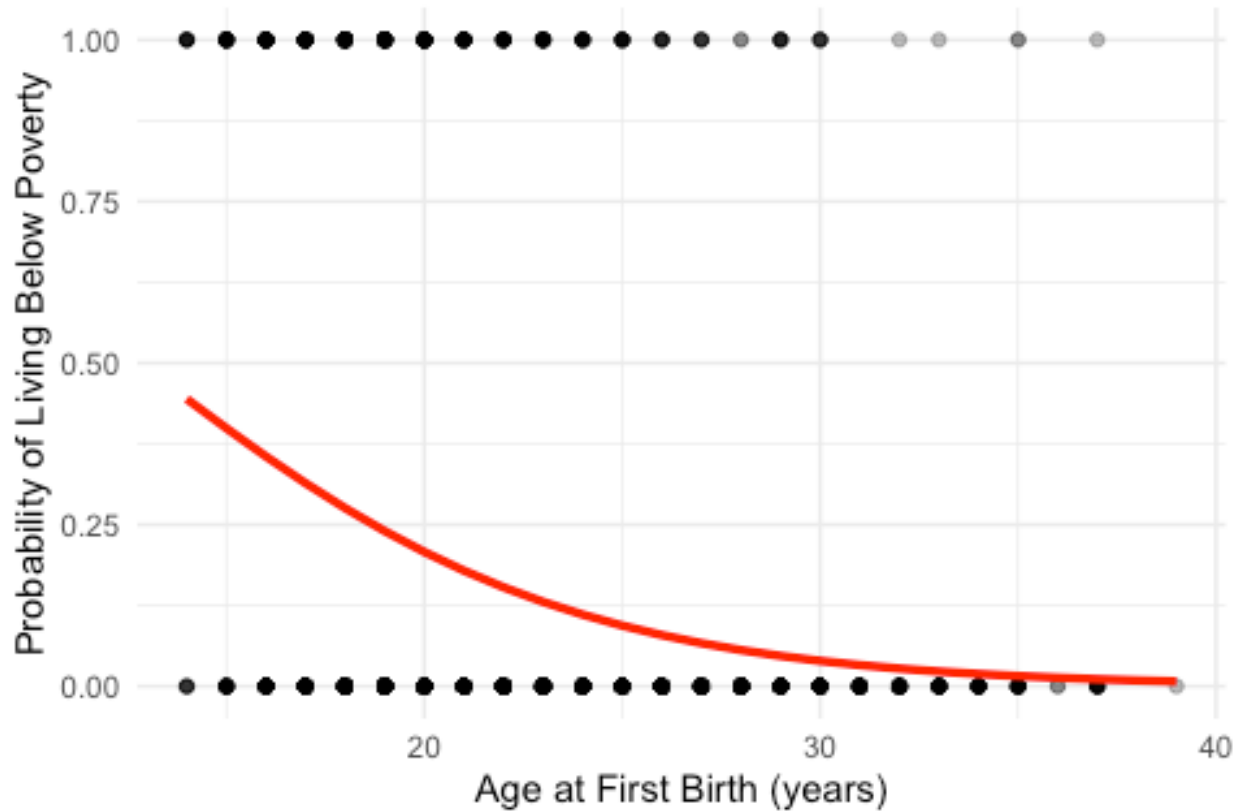
Figure 3. Association Between Age at First Birth and Median Household Income (NHANES)



A simple linear regression model assessed the association between age at first live birth and poverty income ratio. Results indicated that each additional year in age at first birth was associated with a change of 0.12 units in PIR (95% CI: 0.1 to 0.13, $p < 0.05$). The model diagnostics (Residuals vs Fitted and Q-Q plot) suggests that the assumptions are accurate.

A second simple linear regression model examining household income showed that each additional year in age at first birth was associated with a change of approximately 2545.56 USD in median household income (95% CI: 2249.62 to 2841.5).

Figure 4. Predicted Probability of Living Below Poverty by Age at First Birth



Characteristic	OR	95% CI	p-value
Age1stBaby	0.83	0.80, 0.86	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Logistic regression was used to estimate the odds of living below the poverty threshold. Results indicated that age at first birth was associated with reduced odds of poverty (OR = 0.83). The predicted probability plot (Figure 4) demonstrated a decreasing probability of poverty with increasing age at first birth.

Multivariable Analysis

Characteristic	Beta	95% CI	p-value
Age1stBaby	0.03	0.02, 0.05	<0.001
Age	0.02	0.01, 0.02	<0.001
Race1			
Black	—	—	
Hispanic	-0.18	-0.52, 0.16	0.3
Mexican	-0.33	-0.66, 0.00	0.048
White	0.19	-0.03, 0.42	0.090
Other	-0.22	-0.57, 0.13	0.2
MaritalStatus			
Divorced	—	—	
LivePartner	0.14	-0.19, 0.47	0.4
Married	0.81	0.62, 1.0	<0.001
NeverMarried	-0.36	-0.74, 0.02	0.063
Separated	-0.21	-0.60, 0.19	0.3
Widowed	-0.05	-0.31, 0.20	0.7
Education			
8th Grade	—	—	
9 - 11th Grade	0.51	0.17, 0.84	0.003
High School	0.69	0.37, 1.0	<0.001
Some College	1.4	1.0, 1.7	<0.001
College Grad	1.9	1.6, 2.3	<0.001

Abbreviation: CI = Confidence Interval

After adjusting for age, race/ethnicity, marital status, and education, age at first live birth remained significantly associated with poverty income ratio. The adjusted effect estimate was 0.03 (95% CI: 0.02 to 0.05, $p = 0$).

Characteristic	Beta	95% CI	p-value
Age1stBaby	-0.03	-0.07, 0.02	0.3
Race1			
Black	—	—	
Hispanic	-1.1	-2.7, 0.52	0.2
Mexican	-1.8	-3.4, -0.14	0.033
White	-1.1	-2.1, -0.01	0.048
Other	-2.1	-3.9, -0.26	0.025
Age	0.02	0.01, 0.02	<0.001
MaritalStatus			
Divorced	—	—	
LivePartner	0.14	-0.19, 0.47	0.4
Married	0.83	0.64, 1.0	<0.001
NeverMarried	-0.39	-0.77, -0.01	0.046
Separated	-0.18	-0.57, 0.22	0.4
Widowed	-0.03	-0.29, 0.22	0.8
Education			
8th Grade	—	—	
9 - 11th Grade	0.49	0.15, 0.82	0.005
High School	0.68	0.36, 1.0	<0.001
Some College	1.3	1.0, 1.7	<0.001
College Grad	1.9	1.6, 2.3	<0.001
Age1stBaby * Race1			
Age1stBaby * Hispanic	0.05	-0.03, 0.12	0.2
Age1stBaby * Mexican	0.07	-0.01, 0.15	0.083
Age1stBaby * White	0.06	0.01, 0.11	0.017
Age1stBaby * Other	0.09	0.01, 0.16	0.032

Abbreviation: CI = Confidence Interval

An interaction model was fitted to assess whether the association between age at first birth and poverty income ratio differed by race/ethnicity. The interaction term (Age1stBaby x Race1) was evaluated while adjusting for age, marital status, and education (Table 6).

Characteristic	OR	95% CI	p-value
Age1stBaby	0.93	0.89, 0.97	<0.001
Age	0.96	0.95, 0.97	<0.001
Race1			
Black	—	—	
Hispanic	1.22	0.63, 2.35	0.5
Mexican	1.13	0.62, 2.06	0.7
White	0.77	0.49, 1.21	0.2
Other	0.87	0.38, 1.92	0.7
MaritalStatus			
Divorced	—	—	
LivePartner	0.55	0.30, 1.00	0.049
Married	0.28	0.18, 0.43	<0.001
NeverMarried	1.15	0.60, 2.22	0.7
Separated	1.44	0.71, 2.92	0.3
Widowed	0.81	0.48, 1.39	0.5
Education			
8th Grade	—	—	
9 - 11th Grade	0.57	0.32, 1.01	0.052
High School	0.33	0.19, 0.58	<0.001
Some College	0.21	0.12, 0.38	<0.001
College Grad	0.07	0.03, 0.14	<0.001

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

In the adjusted logistic regression model, age at first live birth remained inversely associated with the odds of living below the poverty threshold after controlling for covariates. The adjusted odds ratio for age at first birth was 0.93.

Missing Data & Sensitivity Analysis

The analytic sample derived from the NHANES data set excluded male participants and selected variables relevant to socioeconomic status and reproductive history, including age at first live birth (Age1stBaby), current age, race, poverty income ratio (Poverty), and median household income (HHIncomeMid). Additionally, new variables, age at first live birth decade grouping (Age1stBabyDecade) and poverty status were created to assist in analyses. Missing data was present primarily in the variables Age1stBaby, HHIncomeMid, and Poverty.

After excluding observations with missing values in Age1stBaby, HHIncomeMid, Poverty, and derived variables, a total of 1737 participants remained in the final analytic sample. The exclusion of cases may reduce statistical power and affect the representation of the sample, particularly if the missing values are not random. For example, women who have not had a live birth or who did not report income may differ systematically in socioeconomic status or demographic characteristics.

Missing data in this context is not completely at random, as reproductive history and income reporting may be associated with age, cultural factors, or socioeconomic position. As a result, the analytic sample may over-represent women with complete reproductive histories and economic stability. This can potentially introduce bias into the study causing inaccurate estimations and inferences of the association between age at first birth and socioeconomic status. These limitations should be considered when interpreting the findings as the observed relationships may underestimate or overestimate the true population.

Discussion

Principal Findings and Interpretation

This study evaluated the association between age at first live birth (Age1stBaby) and current socioeconomic status among U.S. women using NHANES 2009-2012 data. Across descriptive, bivariate, and regression analyses, later age at first birth was consistently associated with more favorable socioeconomic outcomes.

Descriptive patterns showed that women with younger ages at first birth (Age1stBabyDecade) had lower poverty income ratios (Poverty) and lower median household income (HHIncomeMid). These trends were supported by both descriptive visualizations and bivariate analyses.

In unadjusted linear regression, each additional year in Age1stBaby was associated with an increase of 0.119 in PIR (95% CI: 0.104 to 0.134) and 2545.56 USD in household income (95% CI: 2249.62 to 2841.5). Although statistically significant, the relatively narrow confidence intervals indicate modest associations with some uncertainty.

Logistic regression results further demonstrated that increasing Age1stBaby was associated with lower odds of living below the poverty threshold (poverty_status) with an odds ratio of 0.83 (95% CI: 0.801 to 0.86). This suggest a protective relationship between delayed childbearing and poverty status.

After adjustment for Age, Race1, MaritalStatus, and Education, the association between Age1stBaby and Poverty remained statistically significant (0.031, 95% CI: 0.015 to 0.047, $p = 1.26 \times 10^{-4}$). Interaction analyses suggested potential variation by race/ethnicity, though there was a wider confidence interval.

Overall, finding support the life course perspective, indicating that earlier reproductive timing is associated with long-term socioeconomic disadvantages, while delayed childbearing is associated with improved economic stability.

Limitations

This study has several limitations. First, this cross-sectional study design limits causal inferences and lacks temporality. Secondly, being that the data was collected from 2009 to 2012, it may accurately depict the current U.S. female population. Third, missing data may also introduce bias if women with incomplete reproductive histories or income data differ systematically from those included in the sample. Due to the nature of this data, the study is particularly vulnerable to selection bias, information bias, and temporal bias. Despite these limitations, this analysis highlights a meaningful link between reproductive timing and socioeconomic outcomes.

Public Health Implications

The timing of first childbirth is more than a personal milestone; it is a turning point that can influence a woman's economic stability for decades. This analysis revealed that women who have their first child at younger ages tend to experience lower socioeconomic status, while those with later first births generally exhibit higher financial stability. These patterns echo the broader life course perspective introduced earlier, illustrating how early reproductive events intertwine with social determinants to shape long-term outcomes (**Levine2009?**; **Moore1993?**).

These findings have important public health implications. Early childbearing can perpetuate cycles of disadvantage, affecting education, income, and access to resources, particularly for populations already facing structural inequities (**Câmara2015?**; **Gigante2019?**).

Interventions aimed at supporting adolescents, promoting reproductive education, and addressing broader social determinants may help reduce long-term socioeconomic disparities. However, limitations must be acknowledged. NHANES data depicts a cross-sectional study, so causal inferences cannot be drawn. In addition, it creates challenges with assessing temporality. Missing data may also introduce bias if women with incomplete reproductive histories or income data differ systematically from those included in the sample. Due to the nature of this data, the study is particularly vulnerable to selection bias, information bias, and temporal bias. Despite these limitations, this analysis highlights a meaningful link between reproductive timing and economic outcomes, reinforcing the value of the life course perspectives in public health research. Building on these findings, future research could examine whether this same positive association exists for male participants.

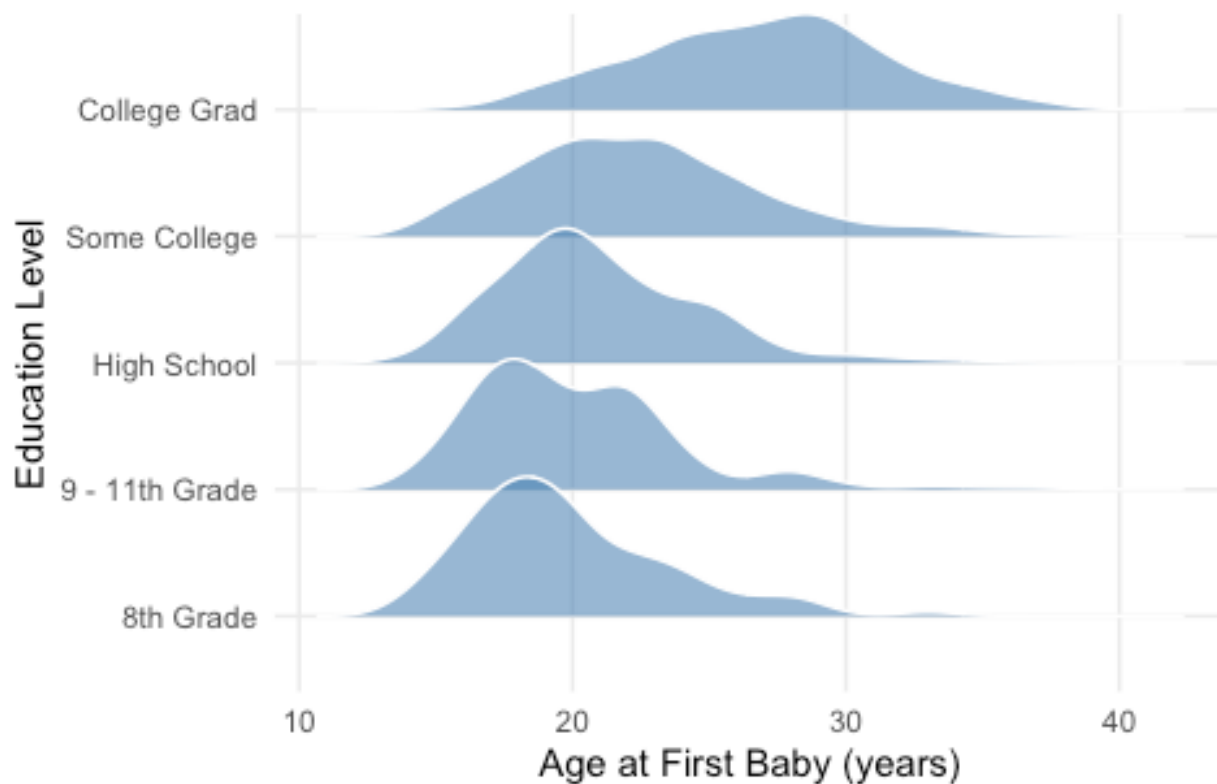
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Appendix

Figure S1. Distribution of Age at First Birth by Education Level (NHANES)



Note. Rigeline plot shows the distribution of age at first birth across education
Education included as a secondary SES indicator.